

Cohort Exercise - Domain & Solution Modeling

Learning Outcomes

By the end of this exercise, you should be able to:

1. Analyse system structure using **Domain Class Diagrams** (Business/Conceptual Phase).
2. Design software architecture using basic **Solution Class Diagrams** (Software/Design Phase).
3. Interpret system execution flow from a given **Sequence Diagram**.

Total 20 Marks

Task 1: The Domain Phase (10 Marks)

Consider the below requirements for a campus food ordering system.

A student signs up for an account in the campus food ordering system. During the sign-up, the student needs to supply the following particulars: 1. the student's email, preferred password, and mobile number. 2. the student's campus delivery details, such as hostel room number or classroom location. 3. the preferred dietary requirements (e.g., Halal, Vegetarian).

Upon successful sign-up, the system sends a confirmation email to the student.

After login, the student browses through the list of food courts inside the campus. The student selects a preferred food court stall to view its available menu items. Upon selecting an item and specifying the delivery location (hostel or classroom), the student needs to pay for the meal that she orders using a credit card. The system verifies the credit card payment. Upon successful payment, the system sends an email confirmation to the student for her order. The system sends an email and SMS notification to the food court stall regarding the new order. When the payment is unsuccessful, the system prompts the student to use a different payment method. The student can logout from the system anytime she wants.

Ten minutes before the estimated delivery time, the system sends an SMS reminder to the student and the delivery rider about the upcoming delivery to the specified classroom or hostel. The food court stall updates the order status to "Preparing", and the rider picks up the order. The student tracks the meal delivery using the student's account.

When the delivery ends, the app prompts the student to leave a food review. The student writes a review and submits the rating to the food court stall through the system. The stall owner acknowledges the student's review and gives a reply and a comment.

Note: For this task, you are mapping the business reality. Do not include software methods, transient dependencies, or memory navigability arrows.

1. Identify at least six domain classes and their attributes in the above scenario.
 2. Identify at least one inheritance, association, aggregation, and composition, and justify your choices.
 3. For the above identified relationships, provide the multiplicities and justify them.
 4. Summarize the above in a UML Domain Class Diagram.
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Task 2: The Solution Phase (5 Marks)

Transition Context: While Task 1 focused on concepts in the real world, this task focuses on the actual software code structure. You will now take the entities from your Domain model and translate them into software classes by adding operations (methods), visibility constraints, and navigability arrows.

Focusing specifically on the **Delivery, Review, and Payment** portions of the narrative from Task 1, draw a basic **Solution Class Diagram**.

Your diagram must include: 1. At least four software classes adapted from your domain entities (e.g., **Order**, **Payment**, **Review**, **Student**). (*Note: Do not formulate controller classes just yet*). 2. Operations/Methods mapped to the appropriate classes (e.g., + **updateStatus()**, + **verify()**). 3. Visibility symbols for attributes (e.g., - for private) and methods (e.g., + for public). 4. Navigability arrows (solid line ->) showing permanent in-memory references (Associations). Variables/Role names should be at the tips of these arrows. 5. **At least one Dependency arrow (dashed line -.->)**. Use this to show a transient software relationship (e.g., when a class uses another class temporarily as a method parameter, such as passing a **CreditCard** object to a **verify()** method). —

Task 3: Software Execution (5 Marks)

Transition Context: While Task 2 shows the static blueprint of the code, a Sequence Diagram shows the code executing dynamically over time in memory.

Given below is a sequence diagram for completing a food delivery.

Identify the chronological order of execution steps. Write your answers exactly in the format: **classA** calls method **m()** in **classB**.

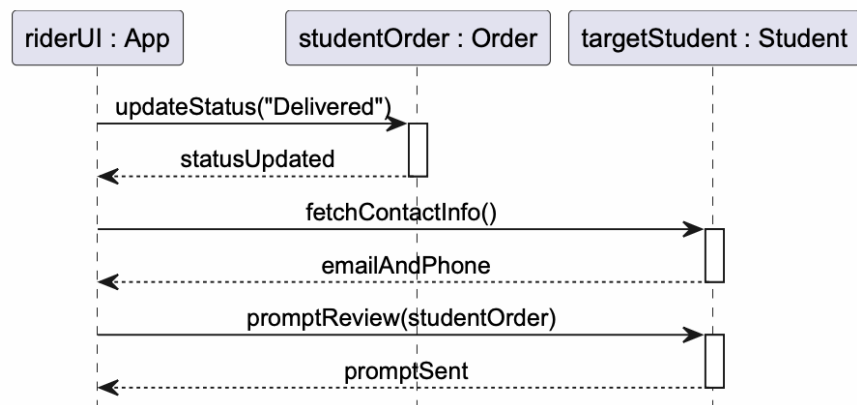


Figure 1: Task 3 Sequence Diagram